Algebra II with Trigonometry Chapter 6.3

Study guide

Saved tutoring response with MathJax rendering. Original PDF file: 4531393_alg-2trig-h-chp-6-3-note.pdf

Certainly! Here is a complete study guide based on the provided document.

Study Guide: Trigonometric Graphs (Chp 6.3)

1. Short Summary

This document introduces the graphs of sine and cosine functions, focusing on their periodic nature, transforms, and how to analyze their equations. It explains amplitude, period, phase shift, and vertical shift for these functions, provides graphical examples, and presents sample problems—some involving real-world applications like sound vibrations.

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2. Core Definitions, Theorems, and Formulas

Definitions

- **Periodic Function:**

A function f is periodic if there is a positive number p such that $f(t+p) = f(t)$ for every t. The smallest such p (if it exists) is called the *period*.

- **Period of Sine and Cosine:**

Both sine and cosine functions have period 2π:

- $\sin(t + 2\pi) = \sin t$

- $\cos(t + 2\pi) = \cos t$

- **Amplitude:**

For $y = a \sin x$ or $y = a \cos x$, the amplitude is $|a|$.

- **Shifted Sine/Cosine Functions:**
- **General Form:**

$y = c + a \sin(k(x - b))$ or $y = c + a \cos(k(x - b))$ where:

- *Amplitude* = $|a|$
- *Period* = $\frac{2\pi}{k}$ for $k > 0$
- *Phase (Horizontal) shift* = b
- *Vertical shift* = c
- One period should be graphed over $[b,\ b + \frac{2\pi}{k}]$

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3. Step-by-Step Study Guide

A. The Basic Sine and Cosine Graphs

- Both are *periodic* with period 2π.
- Key points within one period are where the output is at its maximum (H), minimum (L), and midpoint (M).
- Sine starts at the midpoint (M), goes to high (H), M, low (L), M in one period.
- Cosine starts at high (H), goes to M, L, M, H in one period.

Example Values for Sine:

- $t = 0$: 0 (M)
- $t = \frac{\pi}{2}$: 1 (H)
- $t = \pi$: 0 (M)
- $t = \frac{3\pi}{2}$: -1 (L)
- $t = 2\pi$: 0 (M)

Example Values for Cosine:

- $t = 0$: 1 (H)

- $t = \frac{\pi}{2}$: 0 (M)
- $t = \pi$: -1 (L)
- $t = \frac{3\pi}{2}$: 0 (M)
- $t = 2\pi$: 1 (H)

B. Graphing Transformations

1. **Amplitude:**

- The maximum displacement from the middle (axis). For $y = a \sin x$, it's $|a|$.

2. **Period:**

- Normal period is 2π.
- For $y = a\sin(kx)$ or $y = a\cos(kx)$, period = $\frac{2\pi}{k}$.

3. **Horizontal (Phase) Shift:**

- For $y = a\sin(k(x-b))$ or $y = a\cos(k(x-b))$, graph is shifted right by b.

4. **Vertical Shift:**

- For $y = c + a\sin(k(x-b))$, move graph up by c.

5. **Combined Example:** $y = 3 + \sin 2\pi(x + \frac{1}{8})$

- Amplitude = 1
- Period = 1
- Vertical shift = up 3
- Phase shift = left $\frac{1}{8}$

C. Using Graphing Devices

- Choose appropriate window/view to capture a full period and see the shape.
- State the domain and range:

- Sine and cosine in basic form have ranges $[-a, a]$ (or $[c - a, c + a]$ with vertical shift).

D. Real-World Modeling

- Sine or cosine functions can model periodic phenomena such as sound vibrations.
- *Frequency* = $\frac{1}{\text{period}}$ (number of cycles per second).

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4. Five Practice Problems with Answers

1. Find the amplitude, period, phase shift, and vertical shift of:

$$y = 2 - 3\sin(4x + \pi)$$

Answer:

- Amplitude: 3
- Period: $\frac{2\pi}{4} = \frac{\pi}{2}$
- Phase shift: $-\frac{\pi}{4}$ (left)
- Vertical shift: +2

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2. Write the equation of a cosine function with amplitude 5, period π, and shifted right 2 units.

Answer: Amplitude: 5, period: $\pi \Rightarrow k = \frac{2\pi}{\pi} = 2$ Phase shift: right 2 ⇒ $(x - 2)$ Equation: $y = 5\cos(2(x - 2))$

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3. What is the maximum and minimum value of $y = -4 + 2\sin x$?

Answer: Amplitude = 2, vertical shift -4, so: Max: $-4 + 2 = -2$ Min: $-4 - 2 = -6$

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4. The function $v(t) = 0.5\sin(12\pi t)$ models a vibration. Find the period and frequency.

Answer: Period: $\frac{2\pi}{12\pi} = \frac{1}{6}$ seconds Frequency: $\frac{1}{\frac{1}{6}} = 6$ cycles per second

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5. Over what interval should you graph one period of $y = \sin(5x)$?

Answer: Period is $\frac{2\pi}{5}$, so interval is $[0, \frac{2\pi}{5}]$

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5. Assumptions and Ambiguities

- I inferred that k in $\sin kx$ is always positive, as stated in the notes.
- The standard forms for transformed functions are assumed to be $y = c + a\sin(k(x-b))$ or $y = c + a\cos(k(x-b))$.
- For domain/range, it's assumed basic sine/cosine domains are all real numbers, and the range is determined by amplitude and vertical shift.
- Some example images (like graphs) were referenced but not explicitly described in the text—descriptions were synthesized from context.

End of Study Guide